



demo4-rest-server – JLC Bank -REST API Runs

demo4-rest-server: Files required

1. MyBootApplication.java	2. AccountController.java
3. AccountService.java	4. AccountServiceImpl.java
5. AccountDAO.java	6. Account.java
7. TxInfo.java	8. application.properties
9. pom.xml	

1. MyBootApplication.java

```
package com.jlcindia.springboot;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
public class MyBootApplication {
    public static void main(String args[]) {
        SpringApplication.run(MyBootApplication.class, args);
    }
}
```

2. AccountController.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.DeleteMapping;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.PutMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@RestController
public class AccountController {
```



@Autowired

AccountService accountService;

@PostMapping("/addAccount")

```
public String addAccount(@RequestBody Account acc) {  
    System.out.println("AC-addAccount()");  
    accountService.addAccount(acc);  
    String msg = "Account Added successfully";  
    return msg;  
}
```

@GetMapping("/myaccount/{accno}")

```
public Account getAccountByAccno(@PathVariable("accno") int accno) {  
    System.out.println("AC-getAccountByAccno()" + accno);  
    Account acc = accountService.getAccountByAccno(accno);  
    return acc;  
}
```

@PutMapping("/mydeposit")

```
public void deposit(@RequestBody TxInfo txInfo) {  
    System.out.println("AC-deposit()");  
    accountService.deposit(txInfo);  
}
```

@PutMapping("/mywithdraw")

```
public void withdraw(@RequestBody TxInfo txInfo) {  
    System.out.println("AC-withdraw()");  
    accountService.withdraw(txInfo);  
}
```

@DeleteMapping("/deleteAccount/{accno}")

```
public String deleteAccount(@PathVariable int accno) {  
    System.out.println("AC-deleteAccount()");  
    accountService.deleteAccount(accno);  
    String msg = "Account Deleted successfully";  
    return msg;  
}
```

@GetMapping("/myaccounts")

```
public List<Account> getAllAccounts() {  
    System.out.println("AC-getAllAccounts()");  
    List<Account> accList = accountService.getAllAccounts();  
    return accList;  
}
```



@GetMapping("/myaccounts/city/{city}")

```
public List<Account> getAccountsByCity(@PathVariable String city) {  
    System.out.println("AC-getAllAccounts()");  
    List<Account> accList = accountService.getAccountsByCity(city);  
    return accList;  
}
```

@GetMapping("/myaccounts/bcode/{bcode}")

```
public List<Account> getAccountsByBcode(@PathVariable String bcode) {  
    System.out.println("AC-getAllAccounts()");  
    List<Account> accList = accountService.getAccountsByBcode(bcode);  
    return accList;  
}  
}
```

3. AccountService.java

```
package com.jlcindia.springboot;  
  
import java.util.List;  
/*  
 * @Author : Srinivas Dande  
 * @company : Java Learning Center  
 */  
public interface AccountService {  
    public void deposit(TxInfo txInfo);  
    public void withdraw(TxInfo txInfo);  
    public void deleteAccount(int accno);  
    public void addAccount(Account acc);  
  
    public Account getAccountByAccno(int accno);  
    public List<Account> getAllAccounts();  
    public List<Account> getAccountsByCity(String city);  
    public List<Account> getAccountsByBcode(String bcode);  
}
```

4. AccountServiceImpl.java

```
package com.jlcindia.springboot;  
  
import java.util.List;  
import java.util.Optional;  
import org.springframework.beans.factory.annotation.Autowired;  
import org.springframework.stereotype.Service;  
import org.springframework.transaction.annotation.Transactional;  
/*  
 * @Author : Srinivas Dande  
 * @company : Java Learning Center  
 */
```



@Service

@Transactional

```
public class AccountServiceImpl implements AccountService {
```

@Autowired

```
AccountDAO accountDAO;
```

@Override

```
public void addAccount(Account acc) {  
    System.out.println("ADAO-addAccount()");  
    accountDAO.save(acc);  
}
```

@Override

```
public Account getAccountByAccno(int accno) {  
    System.out.println("ADAO-getAccountByAccno()");  
    Optional<Account> opt=accountDAO.findById(accno);  
    Account acc=null;  
    if(opt.isPresent()) {  
        acc=opt.get();  
    }  
    return acc;  
}
```

@Override

```
public void deposit(TxInfo txInfo) {  
    System.out.println("ADAO-deposit()");  
    Optional<Account> opt=accountDAO.findById(txInfo.getAccno());  
    if(opt.isPresent()) {  
        Account acc=opt.get();  
        double updatedBal=acc.getBalance()+txInfo.getAmount();  
        acc.setBalance(updatedBal);  
        accountDAO.save(acc);  
    }  
}
```

@Override

```
public void withdraw(TxInfo txInfo) {  
    System.out.println("ADAO-withdraw()");  
    Optional<Account> opt=accountDAO.findById(txInfo.getAccno());  
    if(opt.isPresent()) {  
        Account acc=opt.get();  
        double updatedBal=acc.getBalance()-txInfo.getAmount();  
        acc.setBalance(updatedBal);  
        accountDAO.save(acc);  
    }  
}
```



```
@Override
public void deleteAccount(int accno) {
    System.out.println("ADAO-deleteAccount()");
    accountDAO.deleteById(accno);
}

@Override
public List<Account> getAllAccounts() {
    System.out.println("ADAO-getAllAccounts()");
    return accountDAO.findAll();
}

@Override
public List<Account> getAccountsByCity(String city) {
    System.out.println("ADAO-getAccountsByCity()");
    return accountDAO.getAccountsByCity(city);
}

@Override
public List<Account> getAccountsByBcode(String bcode) {
    System.out.println("ADAO-getAccountsByBcode()");
    return accountDAO.getAccountsByBcode(bcode);
}
}
```

5. AccountDAO.java

```
package com.jlcindia.springboot;

import java.util.List;

import org.springframework.data.jpa.repository.JpaRepository;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public interface AccountDAO extends JpaRepository<Account, Integer> {
    public List<Account> getAccountsByCity(String city);
    public List<Account> getAccountsByBcode(String bcode);
}
```



6. Account.java

```
package com.jlcindia.springboot;

import javax.persistence.*;

@Entity
@Table(name = "myaccounts")
public class Account {

    @Id
    @Column(name = "accno")
    @GeneratedValue(strategy = GenerationType.AUTO)
    private int accno;

    @Column(name = "atype")
    private String atype;

    @Column(name = "bcode")
    private String bcode;

    @Column(name = "balance")
    private double balance;

    @Column(name = "city")
    private String city;

    public Account() {}
    public Account( String atype, String bcode, double balance, String city) {
        super();
        this.atype = atype;
        this.bcode = bcode;
        this.balance = balance;
        this.city = city;
    }

    public Account(int accno, String atype, String bcode, double balance, String city) {
        super();
        this.accno = accno;
        this.atype = atype;
        this.bcode = bcode;
        this.balance = balance;
        this.city = city;
    }
    //Setters and Getters
    //toString()
}
```



7. TxInfo.java

```
package com.jlcindia.springboot;

public class TxInfo {
    private int accno;
    private double amount;

    public TxInfo() {}
    public TxInfo(int accno, double amount) {
        super();
        this.accno = accno;
        this.amount = amount;
    }
    //Setters and Getters
}
```

8. application.properties

```
spring.application.name=my-jlc-bank-app
server.port=12345

spring.datasource.driver-class-name=com.mysql.jdbc.Driver
spring.datasource.url=jdbc:mysql://localhost:3306/myjlcdb5
spring.datasource.username=root
spring.datasource.password=srinivas

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update
```

9. pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>demo4-rest-server</artifactId>
<version>1.0</version>
<packaging>jar</packaging>
<name>demo4-rest-server</name>
<description>Spring REST and Data JPA with Boot</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>
```



```
<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.32</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>myjlcapp</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>

</plugins>
</build>
</project>
```

demo4-rest-client – Calling REST API

demo4-rest-client: Files required

1. Account.java	2. TxInfo.java
3. Test1.java	4. Test2.java
5. Test3.java	6. Test4.java
7. Test5.java	8. Test6.java
9. Test7.java	



1. Account.java

```
package com.jlcindia.springboot;

/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class Account {

    private int accno;
    private String atype;
    private String bcode;
    private double balance;
    private String city;

    //Constrcutors
    //Setters and Getters
    //toString()
}
```

2. TxInfo.java

```
package com.jlcindia.springboot;

/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class TxInfo {
    private int accno;
    private double amount;

    //Constrcutors
    //Setters and Getters

}
```



3. Test1.java

```
package com.jlcindia.springboot;

import org.springframework.web.client.RestTemplate;

/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class Test1 {
    public static void main(String[] args) {

        RestTemplate restTemp=new RestTemplate();
        String URL="http://localhost:12345/addAccount";

        Account acc=new Account("SA","B-99",75000,"Hyd");
        String msg=restTemp.postForObject(URL, acc, String.class);
        System.out.println(msg);

        System.out.println("Done");
    }
}
```

4. Test2.java

```
package com.jlcindia.springboot;

import org.springframework.web.client.RestTemplate;

/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class Test2 {
    public static void main(String[] args) {

        RestTemplate restTemp=new RestTemplate();
        String URL="http://localhost:12345/myaccount/101";

        Account acc=restTemp.getForObject(URL, Account.class) ;

        System.out.println(acc);
        System.out.println("Done");

    }
}
```



5. Test3.java

```
package com.jlcindia.springboot;

import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class Test3 {
    public static void main(String[] args) {

        RestTemplate restTemp=new RestTemplate();
String URL="http://localhost:12345/mydeposit";

        TxInfo txInfo=new TxInfo(101, 5000);
restTemp.put(URL,txInfo) ;

        System.out.println("Done");

    }
}
```

6. Test4.java

```
package com.jlcindia.springboot;

import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class Test4 {
    public static void main(String[] args) {

        RestTemplate restTemp=new RestTemplate();
String URL="http://localhost:12345/mywithdraw";

        TxInfo txInfo=new TxInfo(102, 10000);

        restTemp.put(URL,txInfo) ;
        System.out.println("Done");

    }
}
```



7. Test5.java

```
package com.jlcindia.springboot;

import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class Test5 {
    public static void main(String[] args) {
        RestTemplate restTemp = new RestTemplate();
        String URL = "http://localhost:54321/deleteAccount/5";

        restTemp.delete(URL);

        System.out.println("Done");
    }
}
```

8. Test6.java

```
package com.jlcindia.springboot;

import java.util.*;
import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class Test6 {
    public static void main(String[] args) {

        RestTemplate restTemp = new RestTemplate();
        String URL = "http://localhost:54321/myaccounts";

        List<Map<String, String>> accList = restTemp.getForObject(URL, List.class);
        for (Map<String, String> mymap : accList) {
            System.out.println(mymap);
        }

        System.out.println("Done");
    }
}
```



9. Test7.java

```
package com.jlcindia.springboot;

import java.util.List;
import java.util.Map;

import org.springframework.http.ResponseEntity;
import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class Test7 {
    public static void main(String[] args) {

        RestTemplate restTemp = new RestTemplate();
        String URL = "http://localhost:54321/myaccounts/city/Blore";

        ResponseEntity<List> respEntity = restTemp.getForEntity(URL, List.class);

        List<Map<String, String>> accList = respEntity.getBody();
        for (Map<String, String> mymap : accList) {
            System.out.println(mymap);
        }

        System.out.println("Done");
    }
}
```



demo5-rest-server – JLC Bank -REST API Runs

demo5-rest-server: Files required

1. MyBootApplication.java	2. Customer.java
3. CustomerController.java	4. CustomerService.java
5. CustomerIDNotFoundException.java	6. InvalidPhoneNumberException.java
7. CityNotFoundException.java	8. MyErrorMessage.java
9. application.properties	10. pom.xml

1. MyBootApplication.java

```
package com.jlclndia.springboot;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
public class MyBootApplication {
    public static void main(String args[]) {
        SpringApplication.run(MyBootApplication.class, args);
    }
}
```

2. Customer.java

```
package com.jlclndia.springboot;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class Customer {
    private int cid;
    private String cname;
    private String email;
    private long phone;
    private String city;

    //Constrcutors
    //Setters and Getters

    @Override
    public String toString() {
        return "Customer [" + cid + ", " + cname + ", " + email + ", " + phone + ", " + city + "];"
    }
}
```



3. CustomerController.java

```
package com.jlcindia.springboot;

import java.time.LocalDateTime;
import java.util.ArrayList;
import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.http.HttpHeaders;
import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.server.ResponseStatusException;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

@RestController
public class CustomerController {

    @Autowired
    CustomerService customerService;

    @GetMapping("/mycustomer/custId/{custId}")
    public Customer getCustomerByCid(@PathVariable("custId") int custId) {
        System.out.println("CC-getCustomerByCid() - " + custId);

        Customer cust = null;

        try {
            cust = customerService.getCustomerByCid(custId);
        } catch (CustomerIDNotFoundException ex) {
            throw new ResponseStatusException(HttpStatus.NOT_FOUND, "MYCustomer ID Not Found", ex);
        }

        return cust;
    }

    @GetMapping(value = "/mycustomer/phone/{phone}")
    public ResponseEntity<?> getCustomerByPhone(@PathVariable("phone") long phone) {
        System.out.println("CC-getCustomerByPhone() - " + phone);

        HttpHeaders respHeaders = new HttpHeaders();
        Customer cust = null;
```



```
try {
    cust = customerService.getCustomerByPhone(phone);
    respHeaders.add("message", "Phone Number Found");
    return new ResponseEntity<>(cust, respHeaders, HttpStatus.OK);

} catch (InvalidPhoneNumberException ex) {
    respHeaders.add("message", "Phone Number Not Found");
}

MyErrorMessage errMsg = new MyErrorMessage();
errMsg.setMytimestamp(LocalDateTime.now());
errMsg.setMyerror(" Phone Number Not Found");
errMsg.setMystatus(HttpStatus.FORBIDDEN.value());

return new ResponseEntity<>(errMsg, respHeaders, HttpStatus.FORBIDDEN);

}

@GetMapping(value = "/mycustomer/city/{city}")
public ResponseEntity<?> getCustomersByCity(@PathVariable("city") String city) {
    System.out.println("CC-getCustomersByCity() - " + city);

    HttpHeaders respHeaders = new HttpHeaders();
    List<Customer> custList = new ArrayList<>();

    try {
        custList = customerService.getCustomersByCity(city);
        respHeaders.add("message", "Customers Found");
        return new ResponseEntity<>(custList, respHeaders, HttpStatus.OK);
    } catch (CityNotFoundException ex) {
        respHeaders.add("message", "Customers Not Found");
    }

    MyErrorMessage errMsg = new MyErrorMessage();
    errMsg.setMytimestamp(LocalDateTime.now());
    errMsg.setMyerror(" No Customers with given City");
    errMsg.setMystatus(HttpStatus.BAD_REQUEST.value());

    return new ResponseEntity<>(errMsg, HttpStatus.BAD_REQUEST);
}

}
```



4. CustomerService.java

```
package com.jlcindia.springboot;

import java.util.*;
import org.springframework.stereotype.Service;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@Service
public class CustomerService {

    public Customer getCustomerByCid(int custId) {
        Customer cust = null;
        if (custId >= 100) {
            cust = new Customer(custId, "Srinivas", "sri@jlc.com", 12345, "Blore");
        } else {
            throw new CustomerIDNotFoundException();
        }
        return cust;
    }

    public Customer getCustomerByPhone(long phone) {
        Customer cust = null;
        if (phone >= 50000) {
            cust = new Customer(109, "Dande", "dande@jlc.com", phone, "Blore");
        } else {
            throw new InvalidPhoneNumberException();
        }
        return cust;
    }

    public List<Customer> getCustomersByCity(String city) {
        List<Customer> custList = new ArrayList<>();
        if (city.equals("Blore")) {
            custList.add(new Customer(101, "Sri", "sri@jlc.com", 111, city));
            custList.add(new Customer(102, "Vas", "vas@jlc.com", 222, city));
            custList.add(new Customer(103, "SD", "sd@jlc.com", 333, city));
            custList.add(new Customer(104, "DS", "ds@jlc.com", 444, city));
            custList.add(new Customer(105, "Dande", "dande@jlc.com", 555, city));
            custList.add(new Customer(106, "Dande", "dande@jlc.com", 555, city));
        } else {
            throw new CityNotFoundException();
        }
        return custList;
    }
}
```



5. CustomerIDNotFoundException.java

```
package com.jlcindia.springboot;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class CustomerIDNotFoundException extends RuntimeException{

}
```

6. InvalidPhoneNumberException.java

```
package com.jlcindia.springboot;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class InvalidPhoneNumberException extends RuntimeException{

}
```

7. CityNotFoundException.java

```
package com.jlcindia.springboot;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class CityNotFoundException extends RuntimeException {

}
```

8. MyErrorMessage.java

```
package com.jlcindia.springboot;

import com.fasterxml.jackson.annotation.JsonFormat;
import java.time.LocalDateTime;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class MyErrorMessage {
```



```
@JsonFormat(shape = JsonFormat.Shape.STRING, pattern = "yyyy-MM-dd hh:mm:ss")
private LocalDateTime mytimestamp;
private int mystatus;
private String myerror;

public MyErrorMessage() {
}

//Constructors
//Setters and Getters

}
```

9. application.properties

```
spring.application.name=my-jlc-app
server.port=12345
```

10. pom.xml

```
<project ...>
<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.springboot</groupId>
<artifactId>demo5-rest-server</artifactId>
<version>1.0</version>
<packaging>jar</packaging>
<name>demo5-rest-server</name>
<description>Exception Handling- Locally</description>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>
```



```
<build>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>

</plugins>
</build>
</project>
```

demo5-rest-client – Calling REST API

demo5-rest-client: Files required

1. Customer.java	2. Test1.java
3. Test2.java	4. Test3.java

1. Customer.java

```
package com.jlcindia.springboot;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public class Customer {
private int cid;
private String cname;
private String email;
private long phone;
private String city;

//Constrcutors
//Setters and Getters

@Override
public String toString() {
return "Customer [" + cid + ", " + cname + ", " + email + ", " + phone + ", " + city + "];"
}
}
```



2. Test1.java

```
package com.jlcindia.springboot;

import org.springframework.http.HttpEntity;
import org.springframework.http.HttpMethod;
import org.springframework.http.ResponseEntity;
import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class Test1 {

    public static void main(String[] args) {

        RestTemplate restTemp = new RestTemplate();

        String URL1 = "http://localhost:54321/mycustomer/custId/123"; //Success
        String URL2 = "http://localhost:54321/mycustomer/custId/12"; //Failure

        ResponseEntity<Customer> respEntity=null;
        try {
            respEntity = restTemp.exchange(URL2,HttpMethod.GET, HttpEntity.EMPTY, Customer.class);

            Customer cust = respEntity.getBody();
            System.out.println(cust);

        }catch(Exception ex) {
            System.out.println("1."+ respEntity);
            System.out.println("2."+ ex.getMessage());
            System.out.println("3.-----Customer Id Not Found-----");
        }

        System.out.println("Done");

    }

}
```



4. Test2.java

```
package com.jlcindia.springboot;

import org.springframework.http.HttpEntity;
import org.springframework.http.HttpMethod;
import org.springframework.http.ResponseEntity;
import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class Test2 {

    public static void main(String[] args) {

        RestTemplate restTemp = new RestTemplate();

        String URL1 = "http://localhost:54321/mycustomer/phone/54321"; //Success
        String URL2 = "http://localhost:54321/mycustomer/phone/12345"; //Failure

        ResponseEntity<Customer> respEntity=null;
        try {
            respEntity = restTemp.exchange(URL2,HttpMethod.GET, HttpEntity.EMPTY, Customer.class);

            Customer cust = respEntity.getBody();
            System.out.println(cust);

        }catch(Exception ex) {
            System.out.println("1."+ respEntity);
            System.out.println("2."+ ex.getMessage());
            System.out.println("3.-----Customer Phone Not Available-----");
        }

        System.out.println("Done");

    }

}
```



5. Test3.java

```
package com.jlcindia.springboot;

import java.util.List;
import java.util.Map;

import org.springframework.http.HttpEntity;
import org.springframework.http.HttpMethod;
import org.springframework.http.ResponseEntity;
import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

public class Test3 {

    public static void main(String[] args) {

        RestTemplate restTemp = new RestTemplate();

        String URL1 = "http://localhost:54321/mycustomer/city/Blore"; // Success
        String URL2 = "http://localhost:54321/mycustomer/city/Hyd"; // Failure

        ResponseEntity<List> respEntity=null;
        try {
            respEntity = restTemp.exchange(URL2,HttpMethod.GET, HttpEntity.EMPTY, List.class);

            List<Map<String,String>> mylist = respEntity.getBody() ;
            if (!mylist.isEmpty()) {
                for(Map mymap : mylist)
                    System.out.println(mymap);
            }

        }catch(Exception ex) {
            System.out.println("1."+ respEntity);
            System.out.println("2."+ ex.getMessage());
            System.out.println("3.-----No Customers with Given City-----");
        }

        System.out.println("Done");

    }

}
```